

Exploiting Observability in Optical Networks for AI-driven Real-time Soft Failure Detection and Continual Learning

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Abstract *We propose an observability-enabled AI framework using telemetry for degradation management. The proposed autoencoder-based solution detects anomalies, localizes the soft-failures, automates re-action, and supports continual learning, for automatic model retraining while the network is evolving.*
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Overview

Optical technology plays a crucial role in meeting the growing demands for ultra-high reliability, low latency, and guaranteed Quality of Service (QoS) [1, 2]. The increased use of Intelligent Automation Platforms replaces manual tasks with automated functions, reducing operational costs, improving network performance, and enhancing customer experience through machine-learning-based intelligence. The automation model includes automatic management for end-to-end service monitoring, allowing for the early detection of anomalies and rapid, effective interventions. However, applying automation to optical networks is complex, requiring automatic identification of degradations and proactive corrective actions.

Traditional monitoring techniques, like periodic polling and threshold-based alerts, highlight the need for advanced, real-time monitoring solutions [3, 4]. Recent advancements have introduced streaming telemetry as an approach to network observability. Complementing this, AI/ML techniques, particularly unsupervised models like autoencoders, enhance this paradigm by learning normal behavior from telemetry data (e.g., BER, OSNR, CD) and flagging deviations [5].

Two fundamental limitations persist in the current state of the art. First, most monitoring frameworks do not provide real-time insight: anomaly detection and root-cause analysis are executed as separate, batch-oriented stages, obliging operators to huge alarms processing to localize affected span or node. Second, AI frameworks lack on following the network operation during its evolution.

This work demonstrates an integrated, closed-loop architecture designed to address both the lack of real-time insight, and an AI-based soft-failure detection with continual learning capability. High-frequency telemetry is ingested via Kafka to ensure continuous, fine-grained observability. These telemetry streams are processed in real-time by an autoencoder model, trained to reconstruct normal operating conditions. The adoption of autoencoder helps on providing a valid tool to detect all possible

degradations. Anomalies are flagged as soon as the reconstruction error diverges from expected patterns. Crucially, we employ SHapley Additive exPlanations (SHAP) [6] to interpret the model's output and isolate the features most responsible for the deviation. This enables localization of faults to specific physical elements. The model has been designed to support continual learning, triggering the automatic retrain, when an unknown normal condition is detected.

Innovation and novelty

In this demonstration, an AI-based soft failure detection will be live shown in real-time leveraging a remote optical testbed, composed by heterogeneous commercial devices. Three are the relevant key innovations.

1. the exploitation of the observability paradigm: relying on a Kafka-based telemetry system, the collection of relevant metrics from the optical testbed is enabled.
2. the adoption of an autoencoder model for the AI detection of possible degradation and SHAP technique for the localization. In addition, the model offers the capability to automatically re-align with a novel normal conditions, automatically retraining the model in use.
3. real-time validation over a relevant optical testbed, composed by heterogeneous devices, envisaging the integration of observability and the AI framework for real-time effective soft failure detection.

Relevance to ECOC

The demo consists on a Proof of Concept (PoC) for the exploitation of the observability paradigm and the AI-based soft failure detection applied to an optical network. A relevant testbed with heterogeneous resources will be accessed from remote, including Wave-shaper, filters, PIC-based ROADMs, EDFA amplifiers, commercial transponders/muxponders, IPoWDM boxes and Variable Optical Attenuators (VOAs). This demo is based

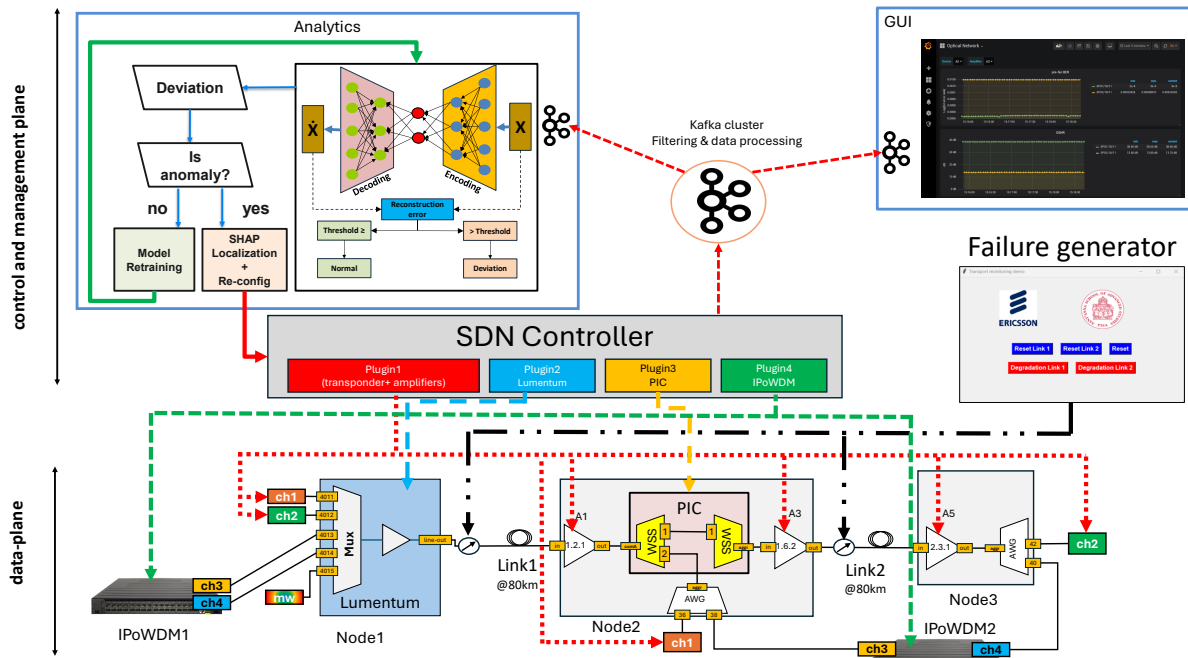


Fig. 1: Reference scenario and system architecture.

on few key points: (i) applying a soft failure detection strategy, based on autoencoder model, to consume in real-time the telemetry data collected from the network, detecting critical conditions, (ii) localizing the degradation point, identifying the cause of the deviation and trying to mitigate the failure effects, (iii) supporting continual learning capability, by automatically retrain the model when a new normal condition (unknown) is detected.

Being present at the ECOC26's demo session will be a relevant opportunity to showcase the level of development of those functionalities, giving the possibility of direct interactions with the audience, triggering discussions with other researchers, with attention to network operators and manufacturers. Interactions and discussions are the key sources to identify possible new research trends, defining new scenarios and/or new requirements.

Demo content and implementation

The demo will be based on the connection to the remote Arno testbed located in Pisa. The access to the testbed will be granted via VPN and dedicated APIs to control the devices. Relevant GUIs and CLIs will be shown live during the demo.

Experimental setup The remote experimental setup for the demonstration is shown in Fig. 1 and highlights a transport segment composed by heterogeneous devices, including three nodes (e.g., Node1, Node2 and Node3). Node1 is implemented with a commercial Lumentum whitebox. Four under test WDM channels (e.g., ch1, ch2, ch3, ch4) are injected in the system, with ch1 and ch2 being traditional WDM muxponder channels at 100Gb/s, while ch3 and ch4 are generated by two coherent pluggable transceivers 400Gb/s ZR+,

installed in an IPoWDM box. Node1 is controlled with an SDN agent based on NETCONF with proprietary data model. Node2 is implemented by combining a commercial Photonic Integrated device (ecploting two PIC filters 1x2) and a set of EDFA amplifiers. An AWG has been adopted to act as DEMUX for the optical signals towards the receivers. Channels ch1 and ch3 are dropped at Node2, traversing a single link of the testbed. Also in this case, a dedicated SDN agent is adopted to control the different components of the node. Node3 is very simple and includes only the AWG to filter the optical signals and an EDFA amplifier. Channels ch2 and ch4 are received at this node, traversing two links of the topology. Another IPoWDM box is used to receive the channels ch2 and ch4 (respectively at Node2 and Node3) of the coherent pluggable transceiver at 400Gb/s.

The two links (e.g., Link1 from Node1 to Node2 and Link2 from Node2 to Node3) are realized with two SMF spools of fiber (80km each). A REST-enabled VOA has been installed at each link to introduce controlled degradation on the signals.

The observability is enforced leveraging a Kafka-based telemetry system while the analytics is based on an autoencoder model, consuming in real time the telemetry data, as depicted in Fig. 1. The telemetry subsystem continuously ingests high-frequency data streams from network elements in a Kafka cluster, ensuring scalability and fault tolerance [4]. The streams include different time-series metrics. For the muxponders and the coherent pluggable transceivers Optical Signal-to-Noise Ratio (OSNR) and Bit Error Rate (BER) are collected, while for the amplifiers aggregated input and output power levels are measured.



Fig. 2: View of the dashboard for real-time monitoring and soft-failure aware topology.

The analytics engine hosts an autoencoder model (trained only with known normal conditions) that reconstructs expected system behavior. Anomalies between actual and reconstructed values are evaluated in real-time. In case of a deviation, the framework analyzes key metrics (BER and OSNR of the existing connections) to distinguish between a new normal condition (new evolution of the network) or a real anomaly. When anomalies are detected, the system leverages SHAP-based attribution to interpret the source of deviation, allowing for rapid soft failure localization. If a new normal condition is identified, the online retraining of the model is performed, to re-synch with the system [7]. Moreover, the metrics, collected from the experimental setup, are displayed in real-time in a Grafana-based GUI, highlighting in real-time both channel performance and a soft-failure aware topology.

Objectives and configuration of the demo The objective of the demo is to show the effectiveness of the AI-based soft-failure detection, showing the real-time behavior with data collected from the experimental setup and reflecting the running configuration. The autoencoder model, used for the soft failure detection, is initially trained offline using a dataset representing normal network operations, collected from the testbed under stable conditions. The telemetry data is a time-series matrix with each line including the selected telemetry parameters (*Input/Output Power of Amplifier 1, Amplifier 3 and Amplifier 5, BER and OSNR of ch1, ch2, ch3 and ch4*). During the execution of the demo the network will evolve with injection of soft failures (on Link1 or on Link2) and new (unknown) normal conditions. The analytics component, based on the autoencoder model, will detect the deviations, detect and localizing degradations or automatically re-synch with the new network state.

Demo presentation to the attendees During the demonstration all the channels will be in operation and soft failures will be introduced on Link1 or on

Link2. The telemetry framework will collect in real-time the data from the testbed. Fig. 2 shows the real-time evolution of OSNR and BER during the Link1 degradation event captured by the monitoring GUI. The data is also consumed by the Analytics module that performs the autoencoder-based analysis, where the reconstruction Mean Squared Error (MSE) is considered. Under normal operation, the MSE remains consistently low. As a soft failure or a new normal condition is introduced, a distinct and rapid increase in the reconstruction error is observed. The MSE value quickly surpasses the anomaly threshold, promptly detecting the deviation from known normal operating conditions. Then, the AI framework performs the metrics analysis, distinguishing among soft failures and new normal condition. If a soft failure is detected, we employ SHAP to understand the cause of detected anomalies. The SHAP analysis explains the reconstruction of the most-deviated feature, localizing the source of the deviation from normal condition. If a new normal condition is detected, the automatic re-training of the model is performed.

The plot in Fig. 2 is an example of closed-loop in case of soft failure on Link1, showing the effects on the involved channels. The system does not perform an instantaneous re-configuration, but wait a few monitoring intervals before the reaction, waiting for the soft failure consistency (avoiding flapping). The plot shows also the subsequent partial recovery, compensating the extra-losses introduced on Link1 by the failure generator. This operation is performed by adapting the gain of the EDFA amplifiers adjacent to the failed link. At the bottom of Fig. 2 the soft-failure aware topology is shown. Also this view is part of the Grafana GUI, highlighting the output of the autoencoder model. In fact, the output of the model is also transmitted via telemetry on a dedicated topic, notifying on the status of the optical network.

Attendees interaction The attendees will actively participate to the demo session. Connectivity services will be running in the remote testbed, exposed APIs on the remote devices will be controlled to introduce soft failures in the testbed. Telemetry data will be impacted by the changes, AI models will identify and localize the degradation, highlighting the relevant aspects on the AI-driven closed-loop reaction. The attendees will have the opportunity to deeply investigate the behavior of the overall system, with particular attention to the key components, including the telemetry framework, the APIs, the AI model analysis and the Grafana webGUI. Live interactions and questions from the attendees will offer the opportunity to deeply investigate the crucial details, offering the possibility to extend the research activities or improve the considered techniques.

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